PRODUCT SPEC — BHANZU MATH MISSION

CANDIDATE	ROLE	COMPANY
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Status: working prototype for the case study. Production integrations are proposed, not built.

1. PROBLEM

About one in three booked demos is not attended, and about 45% of demos booked a week ahead. Between booking and class the family has no reason to re-engage, the child is usually not involved, and joining and scheduling friction appears only at the last minute.

2. USERS

USER	SITUATION	NEEDS
Parent (usually the mother)	Booked from an ad on her phone; busy evenings; wants her child to do well; wary of sales pressure	Remember the class; know what to expect; set the child up without stress; change the time easily
Child, 6–14	Often didn't choose the class; "more maths" can feel like a chore	A reason to be curious; a small win; no sense of being tested
Demo teacher (secondary)	Meets the child cold	A quick sense of the child's level and something to open with
Academic Counsellor (secondary)	Calls the parent after class	A specific, child-centred starting point

3. GOALS AND NON-GOALS

Goals: increase demo attendance (primary); give families a reason to re-engage during the wait (24-hour activation); turn silent no-shows into reschedules; carry engagement into the demo and counsellor call.

Non-goals: assessing or placing the child; replacing the reminder sequence; collecting more personal data than the booking holds; selling during the wait; building a full practice product.

4. REQUIREMENTS

Functional

ID	REQUIREMENT	PROTOTYPE
F1	Home shows child first name, demo date, time **with time zone**, length, teacher, status, live countdown	Built
F2	Countdown handles upcoming, soon (≤30 min), live and past; past state offers a new time	Built
F3	Child onboarding explains the mission (5 puzzles, hints, badge) before choosing	Built
F4	Age bands 6–8, 9–11, 12–14; five puzzles each, easiest first, ending with a mental-math trick	Built
F5	Multiple-choice and typed answers; typed input validated (empty, non-numeric, too long)	Built
F6	Feedback: correct → ✓, praise, explanation; first miss → ×, hint, retry; second miss → answer and explanation	Built
F7	Progress bar and per-puzzle dots update after each resolved puzzle	Built
F8	**Progress survives refresh or leaving**; home offers *Continue mission (n of 5 done)* and *Start over*	Built
F9	Completion: badge, 1–3 stars (never 0), per-puzzle summary, play again, other level, class time	Built
F10	Parent note: skills that needed help, framed for teacher and counsellor; states that nothing is sent	Built (not shared)
F11	Prep page: what to expect before/during/after, tips, FAQ, next steps	Built
F12	Join page: steps, join button (explains it opens 15 min before), readiness checklist, Google Calendar link, copy details, .ics where downloads are allowed	Built (simulated class link)
F13	Reschedule: soonest slots first, 2-hour notice, current slot excluded, choice required, Rescheduled status; original booking time kept for analysis	Built (simulated slots)
F14	Unreadable link data shows a notice and a working sample booking	Built
F15	Reset clears all prototype state	Built
F16	Persistent disclosure that data and integrations are simulated	Built
F17	Event log showing every tracked event with pseudonymous IDs	Built (reviewer tool)
F18	Production: tokenized link per booking, no login, booking from API	Proposed
F19	Production: summary in teacher dashboard and CRM	Proposed
F20	Production: reschedule writes to scheduling and sends a new confirmation	Proposed

Non-functional

- Low-end phones: one static file, about 206 KB uncompressed (65 KB gzip), no login, no install.
- Text ≥16 px; token colour pairs meet WCAG AA 4.5:1 in light and dark themes (checked by `tests/contrast_check.py`).
- Keyboard: native controls; focus moves to each new screen's heading; Escape closes the event log; visible focus outlines.
- Screen readers: feedback in a live region; progress bar exposes its value; ✓/× states also announced in text.
- Works if browser storage is blocked (state lasts for the visit).
- No fake scarcity or pressure; the only countdown is the real class time.

5. USER FLOWS

```
Confirmation message → Home
                       ├→ Start Math Mission → How it works + pick age band → 5 puzzles →
Complete
                       |                                                    |            ├→ Play
again / other level
                       |                                                    |            └→ Get
ready for class
                       |                                          leave/refresh └→ Home: Continue
mission
                       ├→ Get ready for class → How to join
                       ├→ How to join → checklist / calendar / Can't make it? → Change time
                       └→ Change time → pick slot → confirm → Home (Rescheduled)
```

6. ACCEPTANCE CRITERIA (EACH COVERED BY `TESTS/E2E_JOURNEY.PY` UNLESS MARKED)

- Home shows date, time with a zone label, and a ticking countdown for a valid booking.
- A wrong first answer shows a hint, marks the option ×, and does not advance progress.
- A second wrong answer shows the answer and lets the child continue.
- Empty or non-numeric typed answers show an inline error and use no attempt.
- After two answered puzzles, a refresh shows *Continue mission (2 of 5 done)*, which returns to puzzle 3.
- Completion shows the band's badge and stars consistent with the results.
- *Confirm new time* without a choice shows "Choose a time first."; with a choice, home shows Rescheduled, and this persists after refresh.
- A link with an unreadable `demo` value shows a notice and a working sample booking.
- Event log contains the journey events with `booking_id` and `experiment_arm`, and never the child's name.
- Escape closes the event log.
- No horizontal scroll at 390 px (phone) or 768 px (tablet).
- Colour pairs pass 4.5:1 in both themes (`tests/contrast_check.py`).

7. EDGE CASES

CASE	BEHAVIOUR
Demo time passed	"This class time has passed"; status "Time passed"; primary action "Pick a new time"
Class live	"The class is happening now"; join button reads "Join class now"
Same-day booking late at night	Sample demo 3 hours ahead (production uses the real slot)
Unreadable link data	Notice + sample booking
Storage blocked	Works; state lasts for the visit
Corrupted saved progress	Discarded (validated before resuming)

CASE	BEHAVIOUR
Downloads blocked (embedded viewers)	.ics hidden; Google Calendar and copy remain
Copy blocked	Message asks the parent to select the text, which is shown
No open slots	Message explains; production offers a callback
Name with symbols	Only the first name's letters kept

8. PRIVACY

See `docs/ANALYTICS_PLAN.md` §6. In short: first name for display only, age band instead of age, pseudonymous booking ID in analytics, per-question child telemetry is prototype-only pending legal review, no compliance claimed.